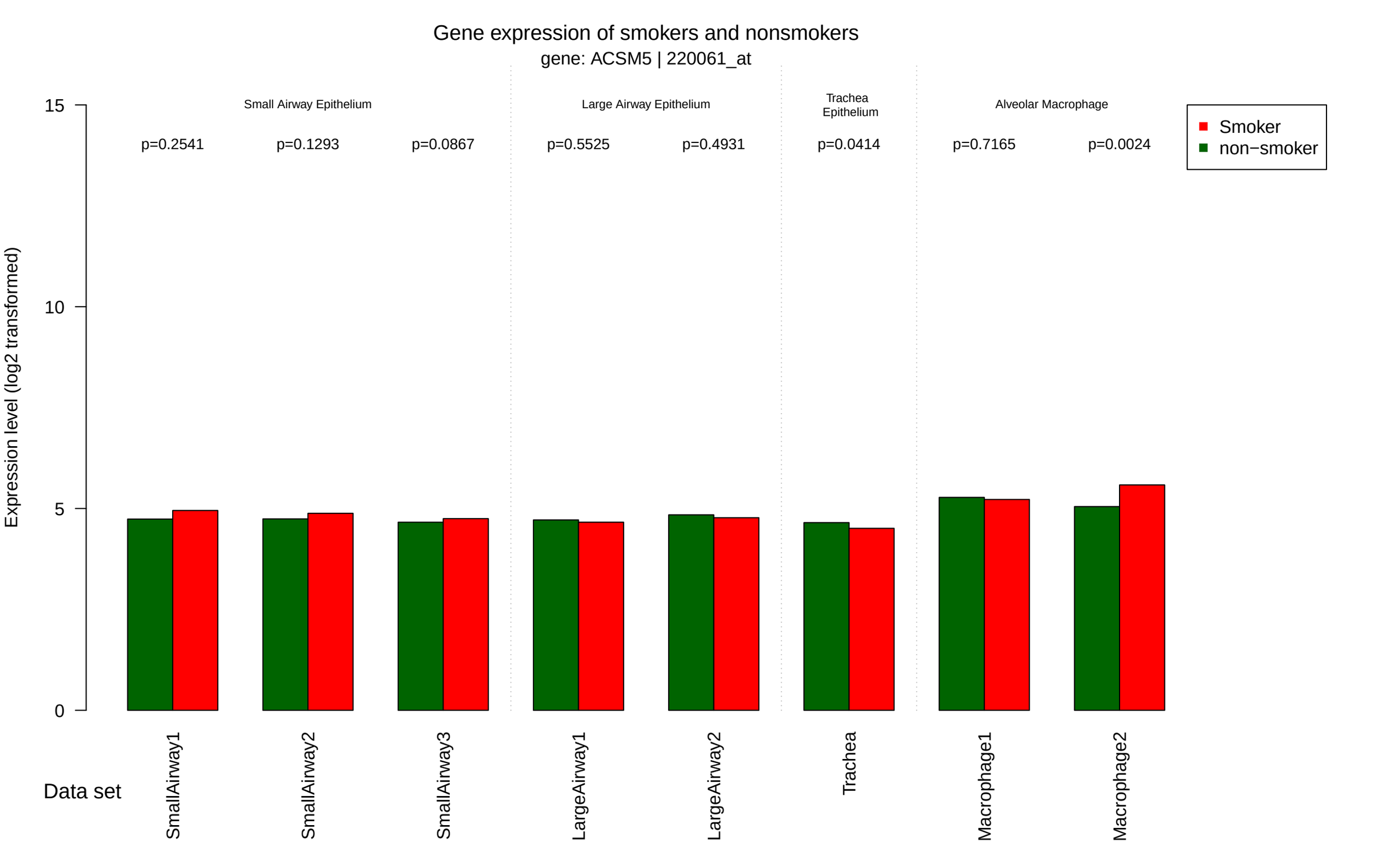
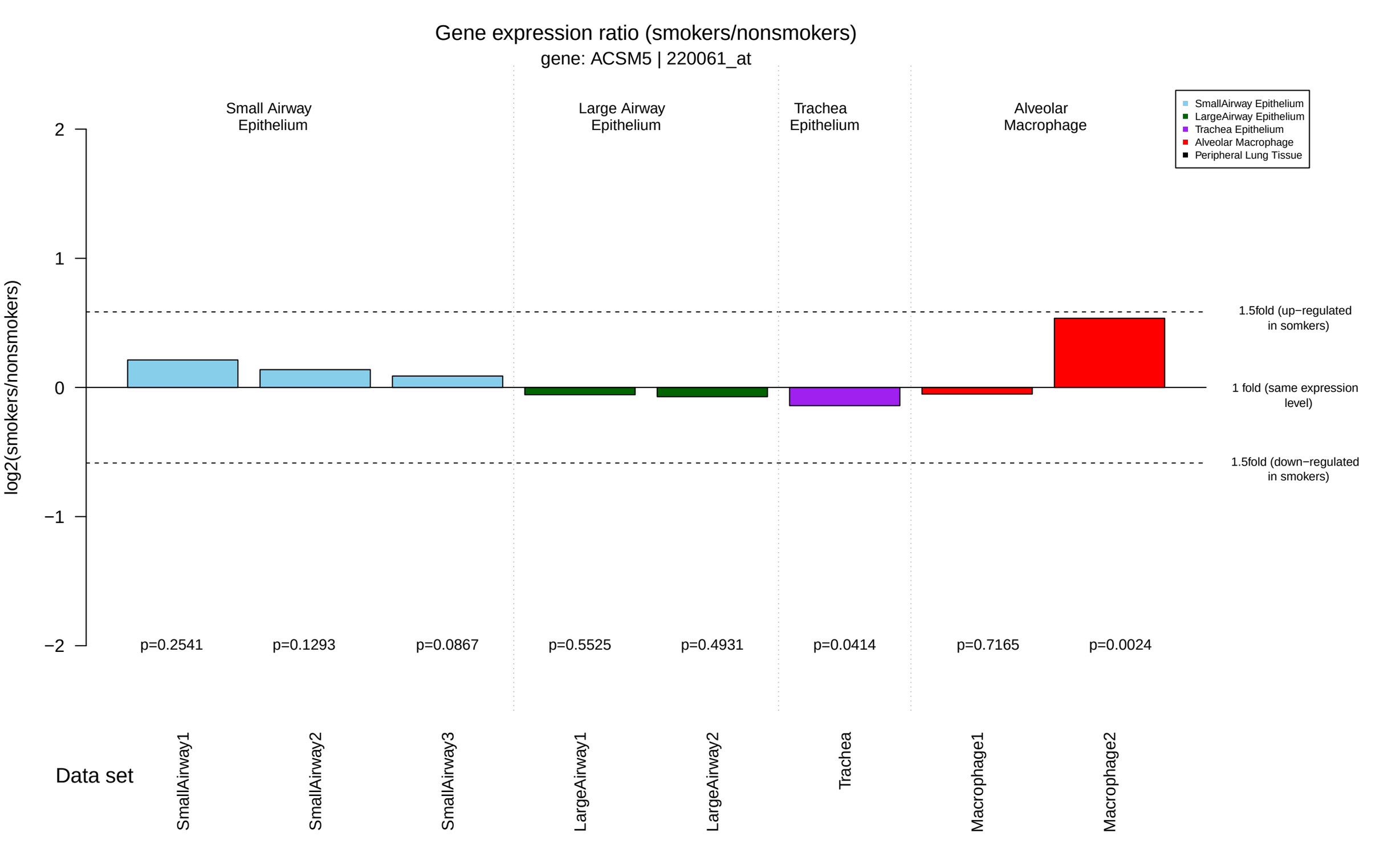
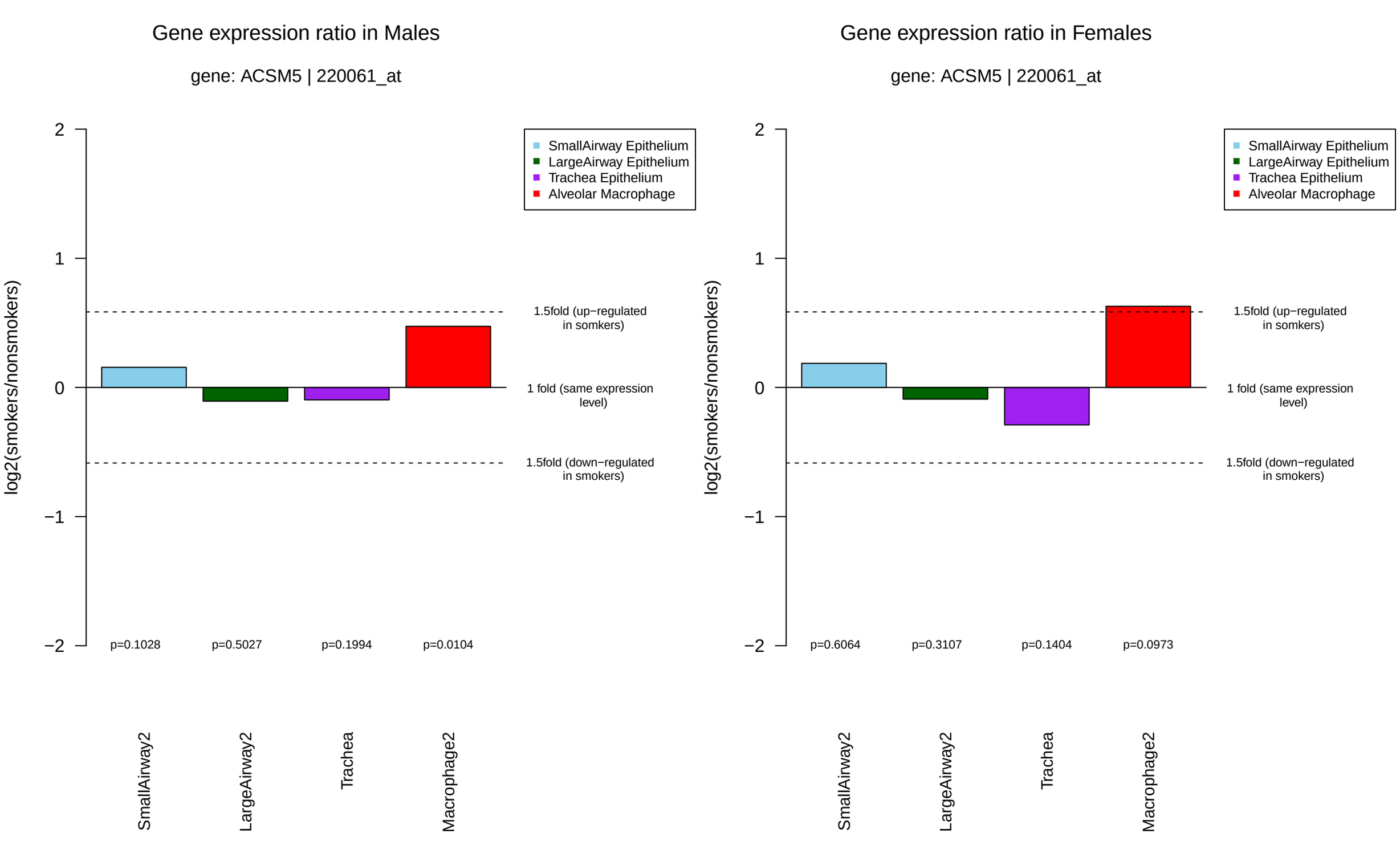
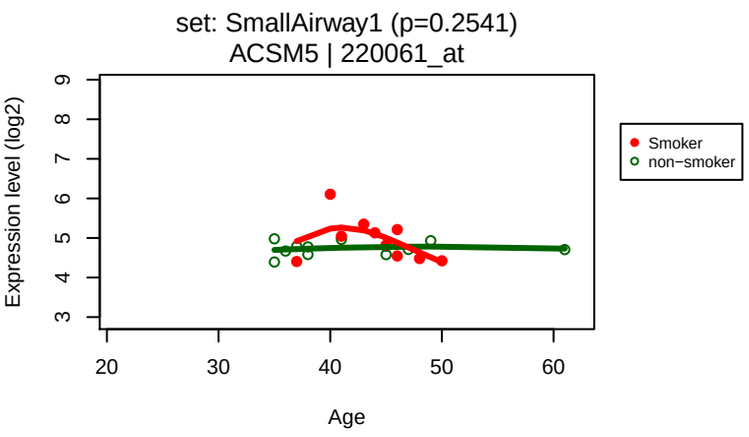
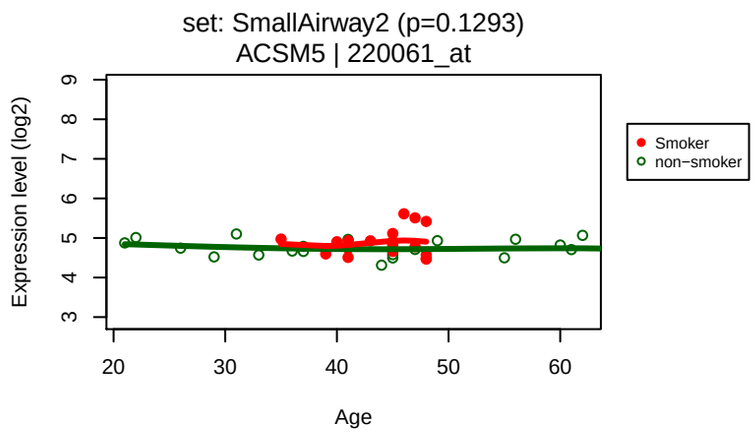
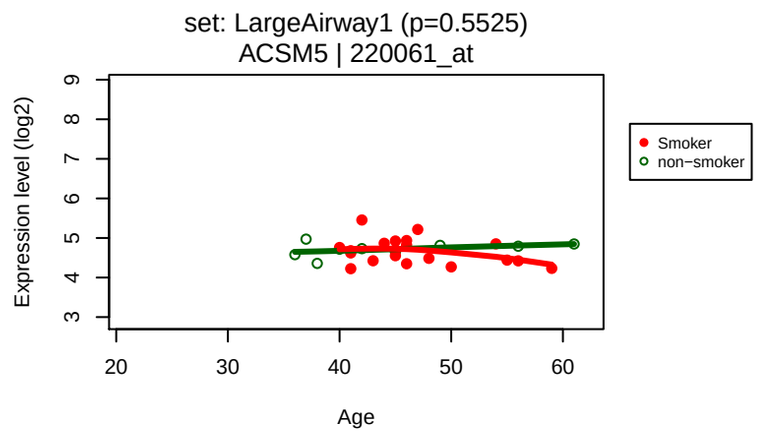

Gene expression of smokers and nonsmokers
gene: ACSM5 | 220061_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2541
p=0.1293
p=0.0867
p=0.5525
p=0.4931
p=0.0414
p=0.7165
p=0.0024
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

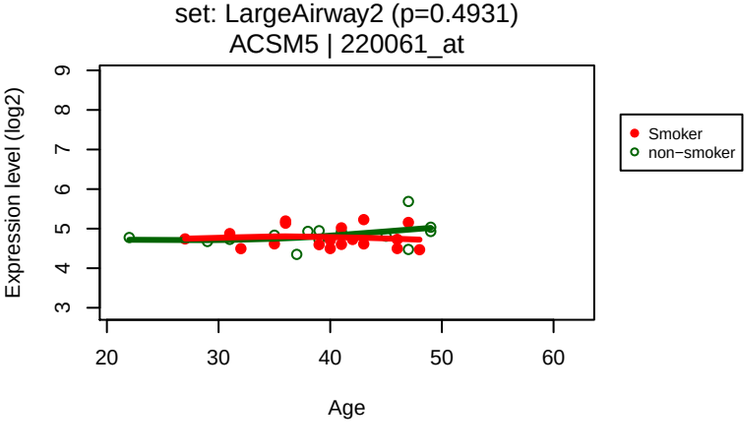

Gene expression ratio (smokers/nonsmokers)
gene: ACSM5 | 220061_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2541
p=0.1293
p=0.0867
p=0.5525
p=0.4931
p=0.0414
p=0.7165
p=0.0024
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

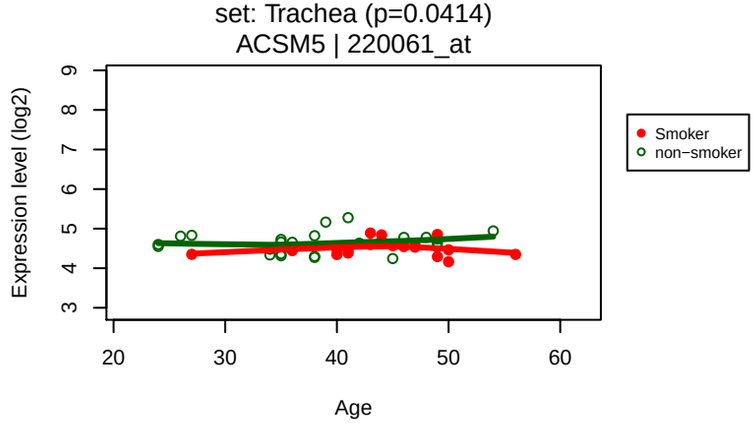

Gene expression ratio in Males
gene: ACSM5 | 220061_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1028
p=0.5027
p=0.1994
p=0.0104
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACSM5 | 220061_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6064
p=0.3107
p=0.1404
p=0.0973
SmallAirway2
LargeAirway2
Trachea
Macrophage2

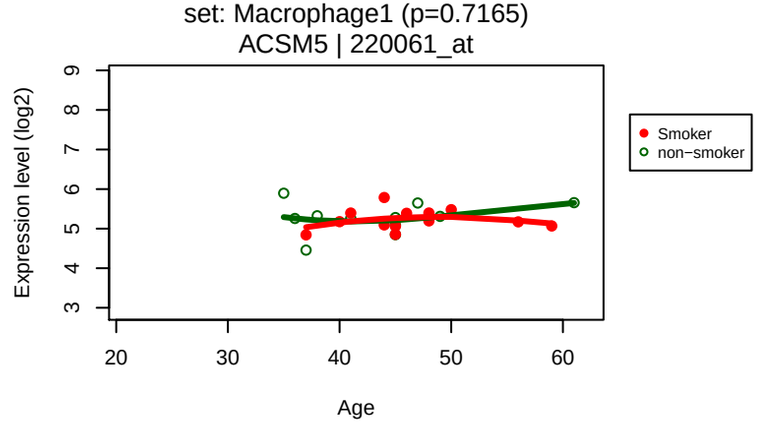

set: SmallAirway1 (p=0.2541)
ACSM5 | 220061_at
Expression level (log2)
Age
Smoker
non−smoker

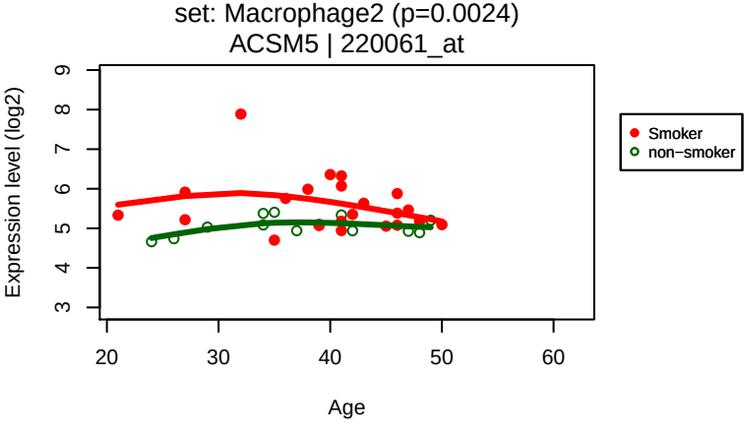

set: SmallAirway2 (p=0.1293)
ACSM5 | 220061_at
Expression level (log2)
Age
Smoker
non−smoker

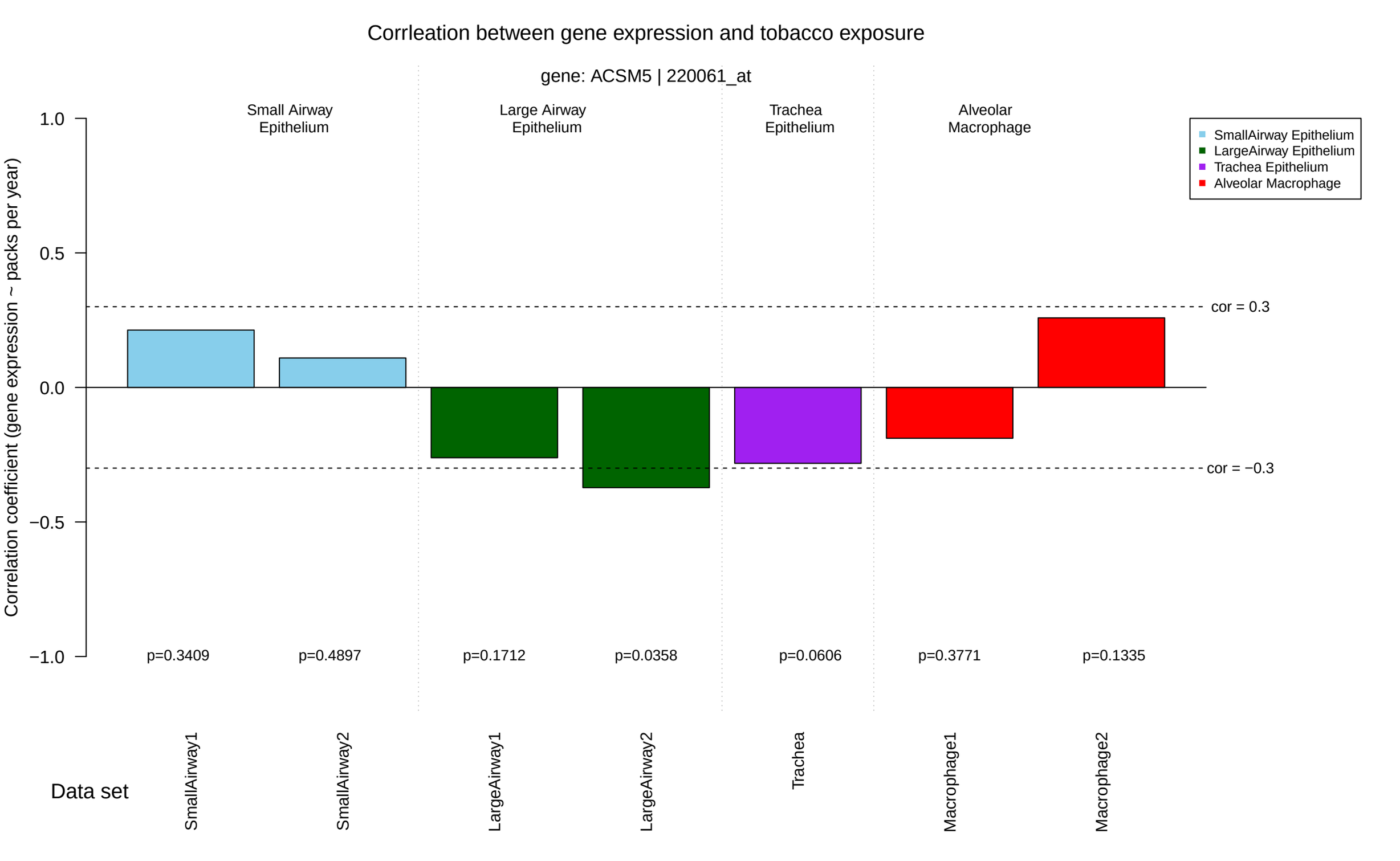

set: LargeAirway1 (p=0.5525)
ACSM5 | 220061_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4931)
ACSM5 | 220061_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0414)
ACSM5 | 220061_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7165)
ACSM5 | 220061_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0024)
ACSM5 | 220061_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACSM5 | 220061_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3409
p=0.4897
p=0.1712
p=0.0358
p=0.0606
p=0.3771
p=0.1335
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2